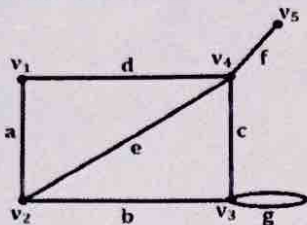
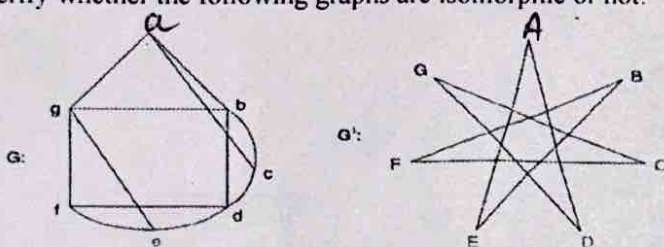
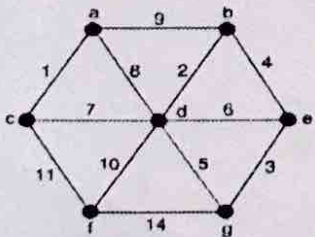




Sub. Code: AI34/AD34/CI35/CY35	Sub.: Discrete Mathematical Structure	Test: I	Date: 18.01.2025
Semester: III	Sections: AI/AD/CI-A, B/CY	Marks: 30	Time: 11 to 12 pm
Term: 03.10.2024 to 25.01.2025			
Note: Answer any TWO full questions. Each main question carries 15 marks.			

Q. No.	Questions	B. L.	CO's	Marks
1. (a)	Define cut-set. Find the cut-set matrix of the following graph 	L3	CO4	5
(b)	Define tree. Prove that a graph G is a tree if and only if it is minimally connected.	L2	CO4	5
(c)	Verify whether the following graphs are isomorphic or not: 	L4	CO3	5
2. (a)	Explain Konigsberg Bridge problem.	L5	CO4	5
(b)	Define (i) Union (ii) intersection (iii) ring sum of two graphs with an example.	L2	CO3	5
(c)	Show that the set Q^+ of all positive rational numbers forms monoid under the composition $*$ defined by $a * b = \frac{ab}{3}$ for all $a, b \in Q^+$.	L3	CO5	5
3. (a)	Define incident matrix. Draw the graph whose incident matrix is $P = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	L3	CO3	5
(b)	Write Kruskal's algorithm to find the shortest spanning tree and hence find the minimal spanning tree of the following graph. 	L4	CO4	5
(c)	Prove that the set of all matrices $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ where a, b are real numbers not both equal to zero form a monoid with respect to matrix multiplication.	L2	CO5	5